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WISP.data package

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Summary

`WISP.data` is an R package designed to automate the acquisition, quality control (QC), analysis, and visualization of spectral data collected by the WISPstation fixed spectroradiometer. Developed by Water Insight as an evolution of the handheld WISP-3 system, the WISPstation is a fixed instrument that records radiance and irradiance across a wavelength range of 350 to 1100 nm (spectral resolution of 4.6 nm). This package serves as a crucial bridge between Water Insight API services and the research community, enabling the transformation of native reflectance measurements into scientifically validated, analysis-ready products through fully reproducible and transparent workflows, from an Open Science perspective.

The package handles the complex transition from total upwelling radiance (L_u), which includes unwanted sky-glint and sun-glint, to the pure water-leaving signal (L_w) by integrating L_{sky} and E_s measurements into standardized atmospheric correction algorithms. `WISP.data` provides a modular set of R functions for: I. downloading data from specific WISPstation by making use of the solutions provided by Water Insight itself (APIs and now via accounts); II. performing automated and reproducible quality control (QC); III. performing automated sky-glint removal (SR); IV. applying different algorithms to spectral measurements; V. visualizing spectral and derived data products.

This makes `WISP.data` an ideal solution for operational water quality monitoring, long-term research applications, and integration into large-scale environmental data pipelines.

Statement of need

The WISPstation is a fixed spectrometer that supports continuous monitoring of water quality; beyond providing high-frequency spectral measurements, it delivers specialized water quality products derived through various algorithms (Gons et al., 1997, 2005; Simis, 2006; van der Woerd & Pasterkamp, 2008), essential for environmental observation, ecosystem assessment, and long-term trend analysis.

The WISPstation operates with 8 specialized channels to optimize data collection:

- **Dual-directional radiance**: two sets of sensors (NNW and NNE) measure upwelling radiance (L_u) and sky radiance (L_{sky}) at 40° angles.
- **Smart orientation**: the system automatically selects the best-oriented sensor set based on the Sun's position, maintaining a relative azimuth angle of approximately 135°.
- **Irradiance and calibration**: includes two downwelling irradiance (E_s) channels and two unexposed "dark" channels to assess sensor degradation over time.

The primary aim of `WISP.data` is the derivation of Remote Sensing Reflectance (Rrs), defined as the ratio between water-leaving radiance (L_w) and downwelling irradiance (E_s):

$$\text{Rrs}(\lambda) = \frac{L_w(\lambda)}{E_s(\lambda)} = \frac{L_u(\lambda) - \rho \cdot L_{sky}(\lambda)}{E_s(\lambda)}$$

Where ρ is the Fresnel reflection coefficient.

However, the effective management and scientific use of these spectral data and derived products present several significant challenges. Data retrieval through API services is often labor-intensive and technically demanding, especially for users without advanced programming experience. In addition, native spectral measurements are sometimes affected by radiometric problems making robust and standardized quality control procedures indispensable. Without rigorous filtering and validation protocols, derived products can be unreliable or scientifically misleading. A further critical barrier lies in the interpretation of spectral signatures themselves. For non-expert users, it is difficult to assess the physical and optical plausibility of reflectance spectra, identify anomalous signals, or distinguish between instrument artifacts and real environmental variability. Moreover, the application of third-party bio-optical algorithms for the estimation of water quality products typically requires substantial domain knowledge, careful parameterization, and consistent preprocessing workflows, which are rarely standardized across studies.

The scientific validity and operational reliability of WISPstation measurements have been demonstrated in several studies, ranging from the detection of climate-driven chlorophyll-a changes during extreme events (Free et al., 2021) to the analysis of phytoplankton spatio-temporal dynamics in Lake Trasimeno (Bresciani et al., 2020). Despite these successful applications, the processing of WISPstation data has been a labor-intensive and time-consuming. Prior to the development of `WISP.data`, researchers often had to manually inspect individual spectral signatures to identify outliers before the data could be used to estimate water quality parameter. This manual quality control process is prone to subjectivity and significantly limits the scalability of high-frequency monitoring.

`WISP.data` addresses these challenges by delivering an integrated, transparent, reproducible, and user-oriented software ecosystem that unifies data acquisition, quality control, spectral validation, and product generation within a single R-based framework. By lowering technical and methodological barriers, the package enables both expert and non-expert users to transform native WISPstation measurements into reliable, scientifically consistent water quality products, fostering reproducibility, comparability, and broader adoption of spectral monitoring technologies in aquatic research and operational monitoring.

State of the field

The current benchmark for processing WISPstation data is WISPcloud (https://wispcloud.waterinsight.nl/) a proprietary web-based infrastructure developed by Water Insight. While WISPcloud provides an efficient 'plug-and-play' solution by delivering ready-to-use products, its 'black-box' nature poses significant challenges for rigorous scientific inquiry. Specifically, the platform lacks the flexibility to modify critical processing parameters, test third-party algorithms, or customize quality control thresholds.

In the R ecosystem, while established packages exist for general hyperspectral analysis (e.g., `hyperSpec` (Beleites et al., 2024)) or for specific in-water optical profilers (e.g., `cops` (Rusch et al., 2021; Rusch & Mair, 2024)), there is a notable software gap for autonomous, fixed, above-water spectroradiometers. Rather than merely replicating existing cloud functionalities, `WISP.data` transforms a proprietary workflow into an open, verifiable, and scientifically controllable ecosystem.

Software design

The design philosophy of `WISP.data` is based on three characteristics: I. a modular, pipeline-oriented workflow abstracting complex API interactions; II. the adoption of community standards for data manipulation and physical units; III. a "transparent-box" approach to quality control, sky-glint removal, and bio-optical modeling. The complete workflow of the `WISP.data` package is shown in Figure 1.

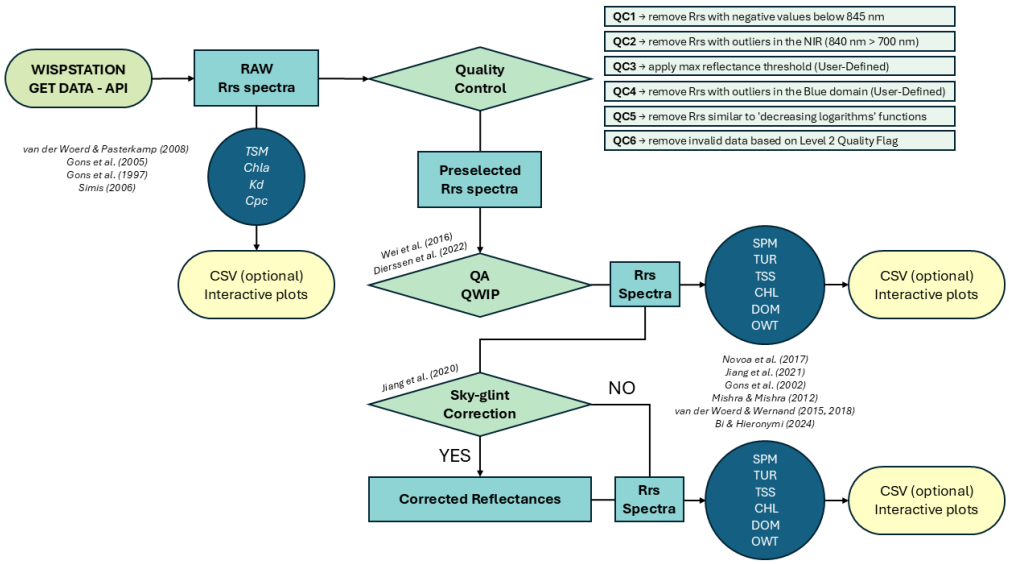

Figure 1. *WISP.data workflow.*

The package architecture is organized into functional modules:

- **Data Integration Layer**: the functions `wisp_get_reflectance_data()` and its iterative wrapper `wisp_get_reflectance_multi_data()` manage the communication with the Water Insight API using the `httr2` framework. Raw JSON/text responses are parsed, unnested, and converted into structured `tibble` objects. This layer also handles the transformation of spectral columns into a standardized format, making the data immediately ready for analysis.
- **Physical Awareness**: `WISP.data` integrates the `units` package to maintain physical consistency. Every measurement retrieved is assigned its proper unit, preventing dimensional errors in downstream calculations.
- **Modular Quality Control (QC)**: the `wisp_qc_reflectance_data()` function implements a multi-step diagnostic pipeline, allowing for a sequential execution of six distinct QC tests (QC1–QC6) alongside advanced validation metrics like QA (Wei et al., 2016) and QWIP (Dierssen et al., 2022). The architecture is highly parameterized, enabling researchers to tune thresholds (e.g., `maxPeak`, `maxPeak_blue`, `qa_threshold`, and `qwip_threshold`) to suit specific optical water types, from oligotrophic to highly turbid environments.
- **Sky-glint removal (SR)**: the `wisp_sr_reflectance_data()` function implements a sky-glint removal algorithm based on the methodology proposed by Jiang et al. (2020).
- **Application of third-party algorithms**: `WISP.data` currently supports a wide array of standardized algorithms for assessing the physical and optical properties of water (Bi & Hieronymi, 2024; Lee et al., 2002; van der Woerd & Wernand, 2015, 2018) and for estimating the concentrations of water quality parameters (Gons, 1999; Gons et al., 2002; Jiang et al., 2021; Mishra & Mishra, 2012; Nechad et al., 2010; Novoa et al., 2017). These are implemented as modular flags within `wisp_qc_reflectance_data()`

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and `wisp_sr_reflectance_data()` functions, allowing for a seamless "one-call" workflow from native spectra to validated products.

- **Visualization layer**: utilizing `ggplot2` and `plotly` as backends, the package provides consistent visualization interfaces. Functions like `wisp_plot_reflectance_data()`, `wisp_plot_comparison()` (Figure 2), and `wisp_trend_plot()` (Figures 3, 4) are designed to handle both publication-quality plots and interactive exploration.

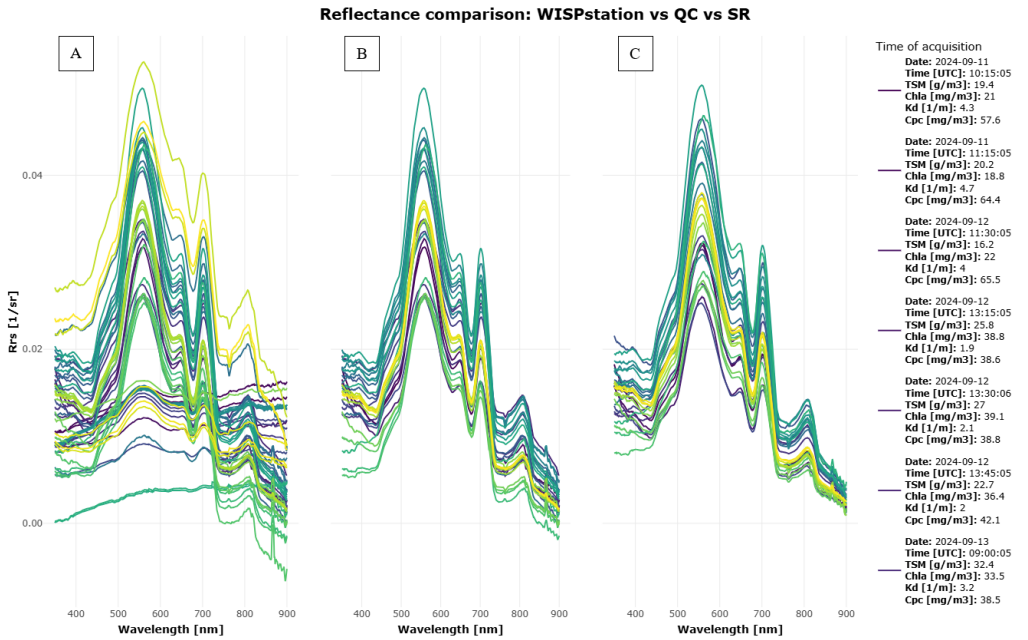

Figure 2. *Plot resulting from the `wisp_plot_comparison()` function showing the comparison between: A) native WISPstation Rrs, B) Rrs filtered by "QC", C) Rrs to which "SR" has been applied (site: Trasimeno; period: 11/09/2024 – 17/09/2024).*

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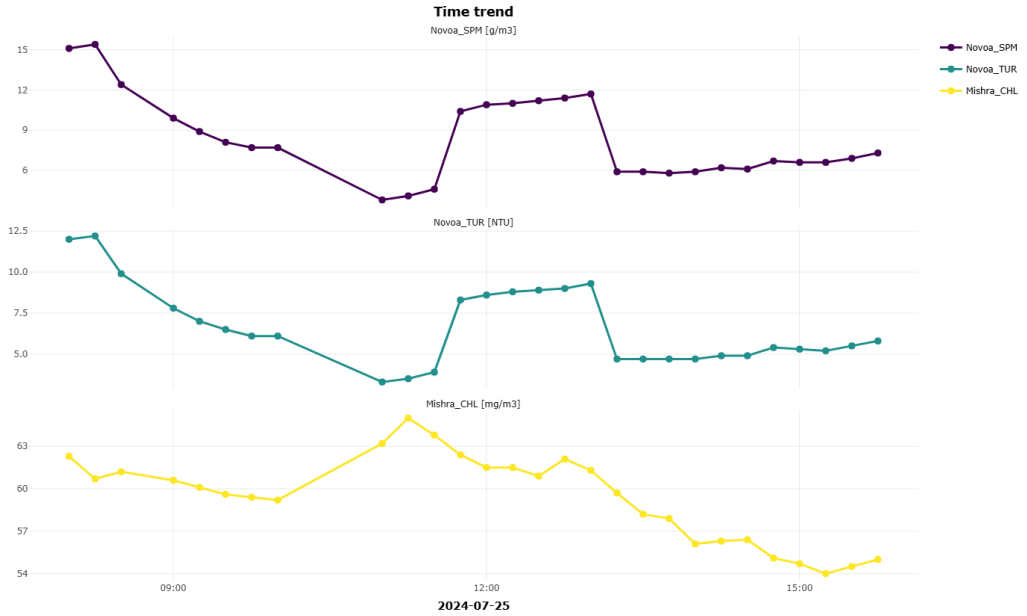

Figure 3. *Plot resulting from the* `wisp_trend_plot()` *function showing the temporal trend of three exemplary parameters (from top to bottom: "Novoa_SPM", "Novoa_TUR", and "Mishra_CHL") for 25/07/2024 from 8 a.m. to 4 p.m. (site: Trasimeno).*

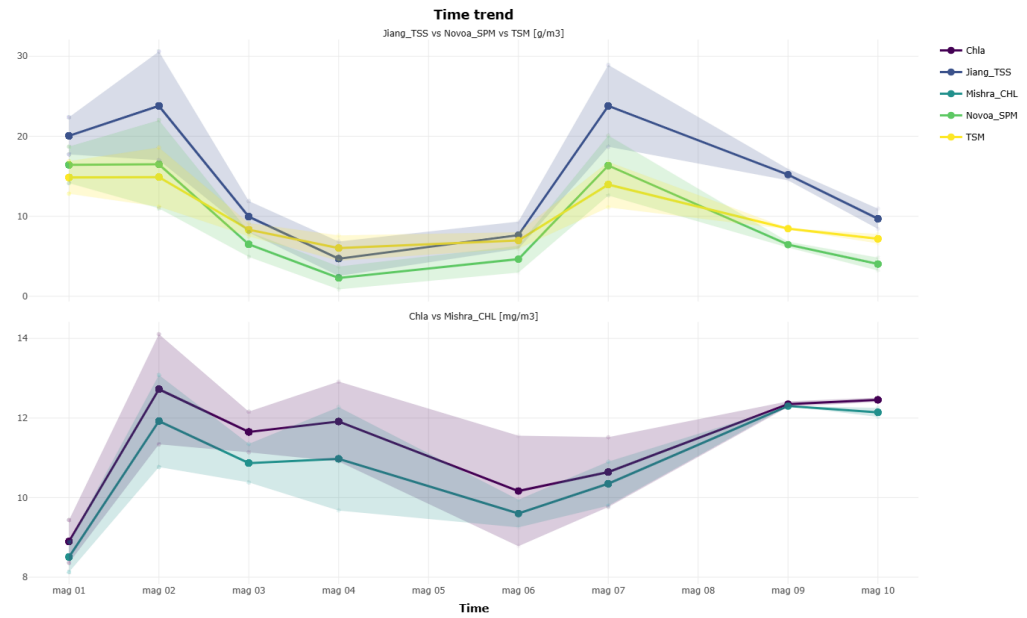

Figure 4. *Plot resulting from the* `wisp_trend_plot()` *function showing the temporal trend of five exemplary parameters averaged on a daily basis: in the upper plot, a comparison between the WISPstation native algorithm (TSM) and two third-party algorithms for estimating suspended solids concentration ("Novoa_SPM" and "Jiang_TSS"); in the lower plot, the comparison between the WISPstation native chlorophyll-a algorithm (Chla) and Mishra_CHL algorithm. (site: Trasimeno; period: 01/05/2024 – 10/05/2024).*

Research impact statement

The package was successfully tested using two WISPstations installed in two different aquatic environments in Italy. The first was installed in April 2018 in Lake Trasimeno, on a lake platform located approximately 400 m from Polvese Island (43.122° N, 12.134° E). The second was deployed in September 2024 along the Po River at Pontelagoscuro (44.889° N, 11.6106° E), approximately 95 km upstream from its confluence with the northern Adriatic Sea, on a floating monitoring platform operated by the Agenzia Interregionale per il fiume Po (AIPO).

Lake Trasimeno, located in central Italy, is the fourth-largest lake in the country (124 km^2). Its catchment is characterized by intensive agricultural practices and livestock farming, which lead to significant nutrient loading. These conditions frequently promote phytoplankton blooms, including potentially harmful cyanobacteria. The lake exhibits complex hydrodynamic behavior, with strong intra-day and inter-day variability driven by meteorological forcing such as temperature fluctuations and wind-induced sediment resuspension. Whereas the Po River, the longest river in Italy, flows across the Po Valley and controls the magnitude and variability of exchanges at the river–sea interface, driving the transfer of sediments, nutrients, and organic matter toward the northern Adriatic Sea and influencing coastal biogeochemistry and water quality. The river is characterized by highly dynamic hydrological conditions, with discharge variability driven by seasonal cycles as well as extreme events such as floods and prolonged low-flow periods. These conditions strongly affect sediment transport and turbidity, resulting in rapid changes in water optical properties.

Both environments have been the focus of long-standing research efforts aimed at understanding the spatio-temporal dynamics of parameters that can also be monitored through remote sensing. They are also included as research sites within two European Research Infrastructures (RIs), namely Integrated European Long-Term Ecosystem, critical zone and socio-ecological Research (eLTER) for Lake Trasimeno[1] (https://deims.org/c93c97fb-196e-480a-87b0-817218cd7c24) and DANUBIUS for the Po River[2].

Within these RIs, the availability of continuous, comparable, quality-controlled, and rigorously validated measurements is essential to minimize uncertainties and ensure data reliability. Such measurements, acquired through high-quality sensor systems and complemented by indicators derived from robust algorithms, can be particularly valuable for the calibration and validation (Cal/Val) of satellite derived products. Within eLTER RI, these measurements contribute to the definition of standardized variables (Zacharias et al., 2026), known as Standard Observations (SOs), which integrate target variables, measurement methods, and protocols to ensure harmonized data collection across sites at the continental scale. Among these, the SO "Phenological traits – SOBIO_15" specifically addresses the use of remote sensing approaches for monitoring vegetation phenological dynamics, including in aquatic ecosystems, encompassing both inland standing waters (lakes) and running waters (rivers).

The scientific utility of `WISP.data` has recently been demonstrated by Ghirardi et al. (under review), who investigated the spatio-temporal dynamics of Suspended Particulate Matter (SPM) and Chlorophyll-a (CHL) in Lake Trasimeno using a synergistic approach. By combining high-frequency in situ observations processed with `WISP.data` and multisensor satellite imagery (2019–2024), the study validated Rrs products from 14 different satellite sensors. The strong agreement between satellite-derived estimates and the `WISP.data` processed ground-truth enabled accurate retrieval of SPM and CHL concentrations. These results highlight the package's role in bridging the gap between local autonomous measurements and large-scale satellite monitoring, facilitating the study of rapid environmental changes in lake ecosystems. For the Po River, `WISP.data` has been used to process high-frequency hyperspectral radiometric data collected at the Pontelagoscuro platform, including quality control, sky-glint correction, and the application of algorithms to derive turbidity from optical measurements. The resulting

[1] https://elter-ri.eu
[2] https://danubius-ri.eu

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time series of above-water observations supports satellite validation and the assessment of atmospheric correction algorithms. In addition, radiometrically derived turbidity was compared with in situ measurements acquired by submerged sensors, enabling consistency assessments between optical and direct observations. This integrated approach enhances the interpretation of turbidity dynamics in highly variable riverine systems.

AI usage disclosure

No generative AI tools were used in the the writing of this manuscript, or the preparation of supporting materials. Generative AI tools were used as a support for code development.

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References

Beleites, C., Bonifacio, A., Dahms, M., Egert, B., Fuller, S., Gegzna, V., Guliev, R., Hanson, B. A., Hermes, M., Kammer, M., Kiselev, R., Mellor, S., Oduniyi, E., & Sergo, V. (2024). *hyperSpec: a package to handle hyperspectral data sets in R*. https://r-hyperspec.github.io/hyperSpec

Bi, S., & Hieronymi, M. (2024). Holistic optical water type classification for ocean, coastal, and inland waters. *Limnology and Oceanography*, *69*(7), 1547–1561. https://doi.org/10.1002/lno.12606

Bresciani, M., Pinardi, M., Free, G., Luciani, G., Ghebrehiwot, S., Laanen, M., Peters, S., Della Bella, V., Padula, R., & Giardino, C. (2020). The Use of Multisource Optical Sensors to Study Phytoplankton Spatio-Temporal Variation in a Shallow Turbid Lake. *Water*, *12*(1), 284. https://doi.org/10.3390/w12010284

Dierssen, H. M., Vandermeulen, R. A., Barnes, B. B., Castagna, A., Knaeps, E., & Vanhellemont, Q. (2022). QWIP: A quantitative metric for quality control of aquatic reflectance spectral shape using the apparent visible wavelength. *Frontiers in Remote Sensing*, *3*, 869611. https://doi.org/10.3389/frsen.2022.869611

Free, G., Bresciani, M., Pinardi, M., Giardino, C., Alikas, K., Kangro, K., Rõõm, E. I., Vaičiūtė, D., Bučas, M., Tiškus, E., Hommersom, A., & Peters, S. (2021). Detecting Climate Driven Changes in Chlorophyll-a Using High Frequency Monitoring: The Impact of the 2019 European Heatwave in Three Contrasting Aquatic Systems. *Sensors*, *21*(18), 6242. https://doi.org/10.3390/s21186242

Gons, H. J. (1999). Optical teledetection of chlorophyll a in turbid inland waters. *Environmental Science & Technology*, *33*(7), 1127–1132. https://doi.org/10.1021/es9809657

Gons, H. J., Ebert, J., & Kromkamp, J. (1997). Optical teledetection of the vertical attenuation coefficient for downward quantum irradiance of photosynthetically available radiation in turbid inland waters. *Aquatic Ecology*, *31*(3), 299–311. https://doi.org/10.1023/A:1009924823521

DRAFT

Gons, H. J., Rijkeboer, M., & Ruddick, K. G. (2002). A chlorophyll-retrieval algorithm for satellite imagery (Medium Resolution Imaging Spectrometer) of inland and coastal waters. *Journal of Plankton Research*, *24*(9), 947–951. https://doi.org/10.1093/plankt/24.9.947

Gons, H. J., Rijkeboer, M., & Ruddick, K. G. (2005). Effect of a waveband shift on chlorophyll retrieval from MERIS imagery of inland and coastal waters. *Journal of Plankton Research*, *27*(1), 125–127. https://doi.org/10.1093/plankt/fbh151

Jiang, D., Matsushita, B., Pahlevan, N., Gurlin, D., Lehmann, M. K., Fichot, C. G., & O'Donnell, D. (2021). Remotely estimating total suspended solids concentration in clear to extremely turbid waters using a novel semi-analytical method. *Remote Sensing of Environment*, *258*, 112386. https://doi.org/10.1016/j.rse.2021.112386

Jiang, D., Matsushita, B., & Yang, W. (2020). A simple and effective method for removing residual reflected skylight in above-water remote sensing reflectance measurements. *ISPRS Journal of Photogrammetry and Remote Sensing*, *165*, 16–27. https://doi.org/10.1016/j.isprsjprs.2020.05.003

Lee, Z., Carder, K. L., & Arnone, R. A. (2002). Deriving inherent optical properties from water color: a multiband quasi-analytical algorithm for optically deep waters. *Applied Optics*, *41*(27), 5755–5772. https://doi.org/10.1364/AO.41.005755

Mishra, S., & Mishra, D. R. (2012). Normalized difference chlorophyll index: A novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sensing of Environment*, *117*, 394–406. https://doi.org/10.1016/j.rse.2011.10.016

Nechad, B., Ruddick, K. G., & Park, Y. (2010). Calibration and validation of a generic multisensor algorithm for mapping of total suspended matter in turbid waters. *Remote Sensing of Environment*, *114*(4), 854–866. https://doi.org/10.1016/j.rse.2009.11.022

Novoa, S., Doxaran, D., Ody, A., Vanhellemont, Q., Lafon, V., Lubac, B., & Gernez, P. (2017). Atmospheric corrections and multi-conditional algorithm for multi-sensor remote sensing of suspended particulate matter in low-to-high turbidity levels coastal waters. *Remote Sensing*, *9*(1), 61. https://doi.org/10.3390/rs9010061

Rusch, T., & Mair, P. (2024). *cops: Cluster Optimized Proximity Scaling*. https://doi.org/10.32614/CRAN.package.cops

Rusch, T., Mair, P., & Hornik, K. (2021). Cluster Optimized Proximity Scaling. *Journal of Computational and Graphical Statistics*, *30*(4), 1156–1167. https://doi.org/10.1080/10618600.2020.1869027

Simis, S. G. H. (2006). *Blue-green catastrophe: Remote sensing of mass viral lysis of cyanobacteria* [PhD thesis, Vrije Universiteit Amsterdam]. https://scholarlypublications.universiteitleiden.nl/handle/1887/11019

van der Woerd, H. J., & Pasterkamp, R. (2008). HYDROPT: A fast and flexible method to retrieve chlorophyll-a from multispectral satellite observations of optically complex coastal waters. *Remote Sensing of Environment*, *112*(4), 1795–1807. https://doi.org/10.1016/j.rse.2007.09.001

van der Woerd, H. J., & Wernand, M. R. (2015). True colour classification of natural waters with medium-spectral resolution satellites: SeaWiFS, MODIS, MERIS and OLCI. *Sensors*, *15*(10), 25663–25680. https://doi.org/10.3390/s151025663

van der Woerd, H. J., & Wernand, M. R. (2018). Hue-angle product for low to medium spatial resolution optical satellite sensors. *Remote Sensing*, *10*(2), 180. https://doi.org/10.3390/rs10020180

Wei, J., Lee, Z., & Shang, S. (2016). A system to measure the data quality of spectral remote-sensing reflectance of aquatic environments. *Journal of Geophysical Research: Oceans*, *121*(11), 8189–8207. https://doi.org/10.1002/2016JC012126

DRAFT

Zacharias, S., Lumpi, T., Weldon, J., Dirnboeck, T., Gaillardet, J., Haase, P., & Mirtl, M. (2026). Achieving harmonized and integrated long-term environmental observation of essential ecosystem variables – elter's framework of standard observations. *Earth's Future*. https://doi.org/10.1029/2025EF006743

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